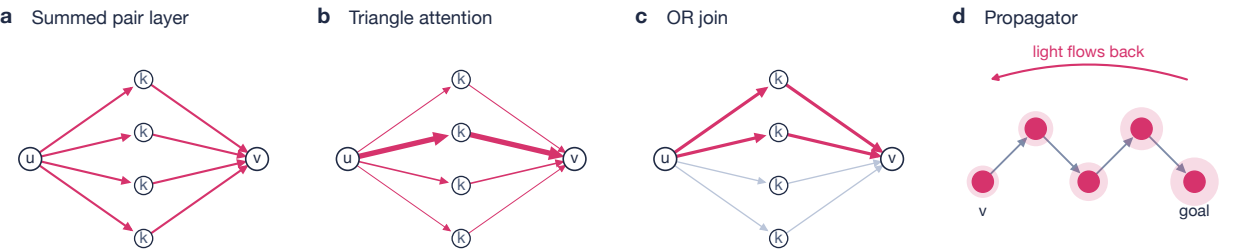

$\frac{1}{\sqrt{n}} \sum_k a_{uk} \odot b_{kv}$

every middle town counts equally; a signed count

$\sum_k \mathrm{softmax}_k(\cdot)\, a_{uk} \odot b_{kv}$

a softmax over the middle town; its score couples both legs

$1 - \exp(-\sum_k \sigma_{uk}\, \sigma_{kv})$

is there at least one middle town? one or many give about 1

$X = S(I - G)^{-1}$

walks of every length at once, in one linear solve